

# PD\_SMARTKEY\_EMN-EVAN\_V2

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| 1    | COVER PAGE       |
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| 3    | VOLTAGE SOURCE   |
| 4    | MICRO CONTROLLER |

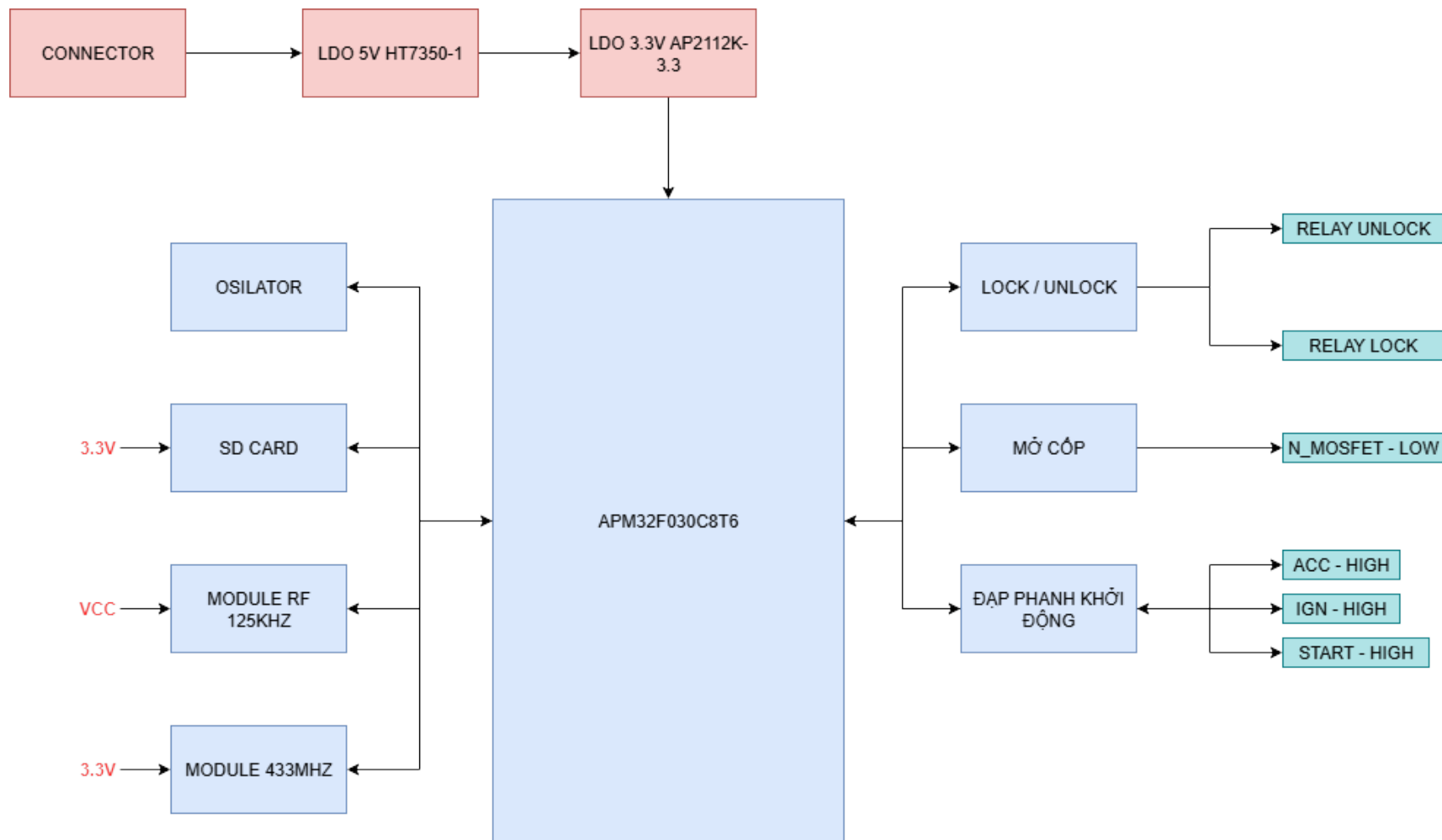
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## DESIGN CONSIDERATIONS

**DESIGN NOTE:**  
Functions of this block

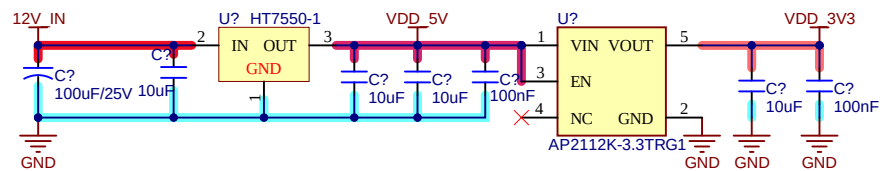
**DESIGN NOTE:**  
Calculation equations

**DESIGN NOTE:**  
Calculations when choosing this component



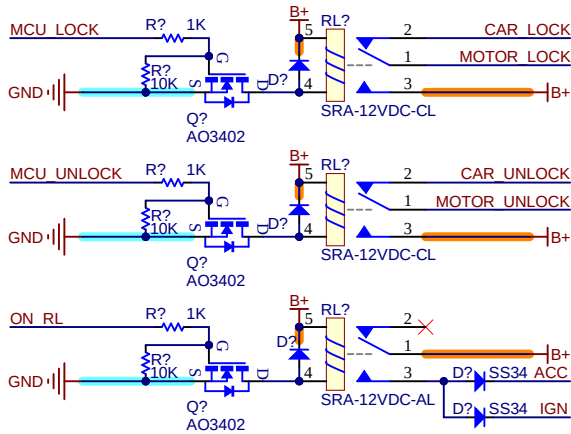
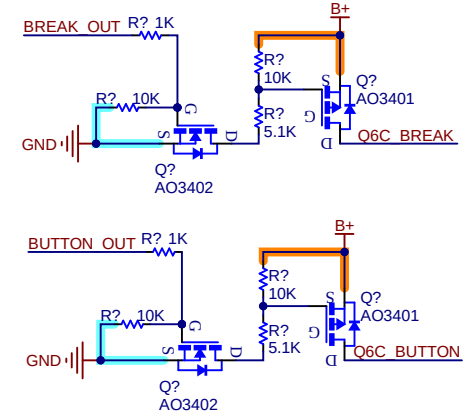
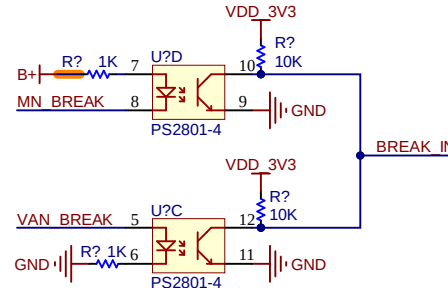
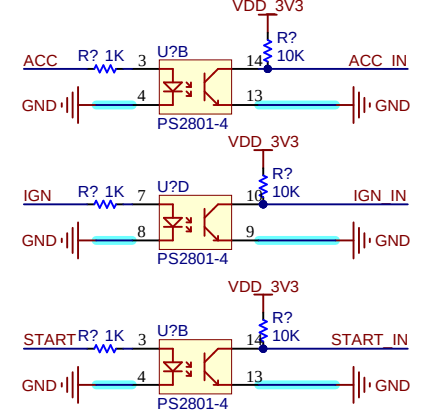
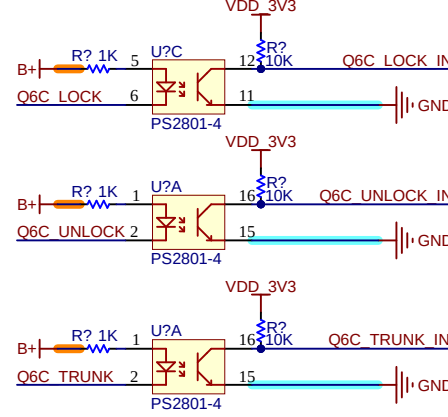
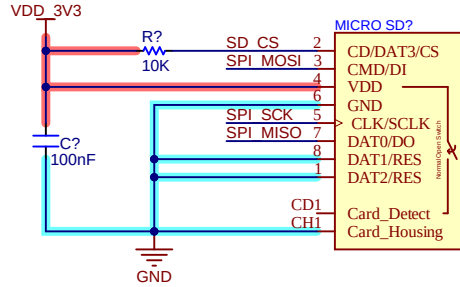
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|                 |            |              | Approved: *              |
|                 |            |              | <b>CONFIDENTIAL</b>      |
| Size: A4        | Project:   |              | Modified Date: 8/10/2026 |
| Date: 8/10/2026 | 5:24:35 PM | Sheet 2 of 6 | Revision: *              |
|                 |            |              | SW Version: 24.1.2.44    |





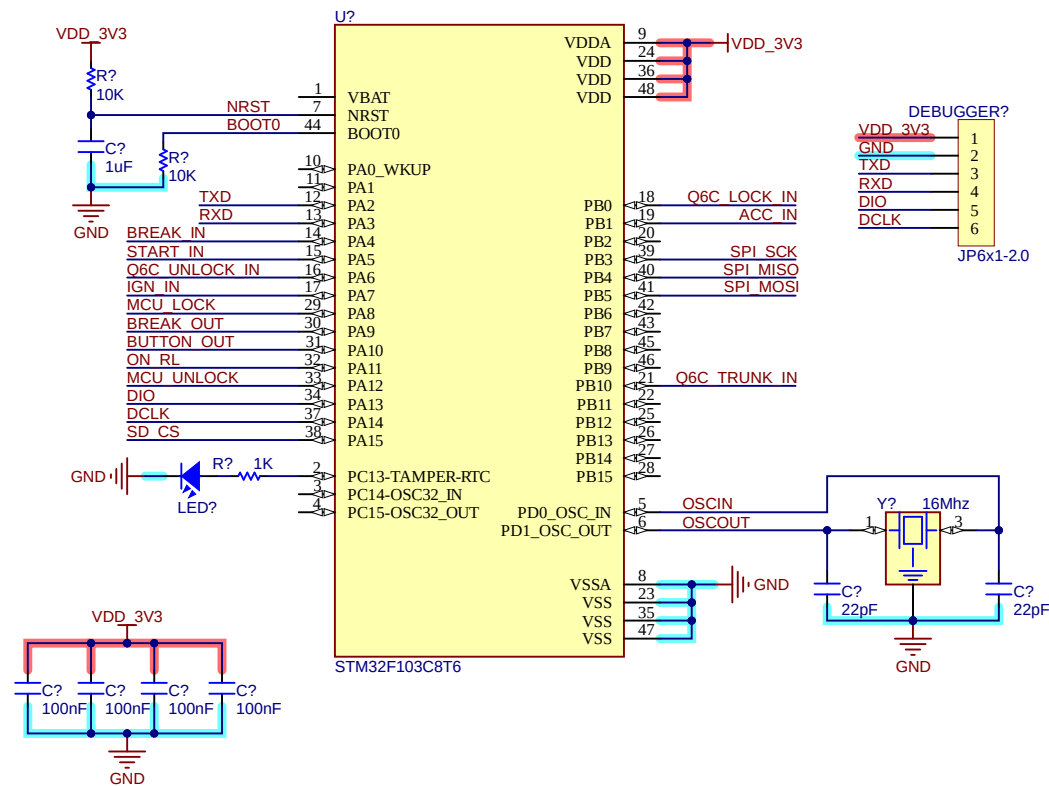
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| 5:24:36 PM      |  |  | Revision: *              |
| Sheet 3 of 6    |  |  | SW Version: 24.1.2.44    |





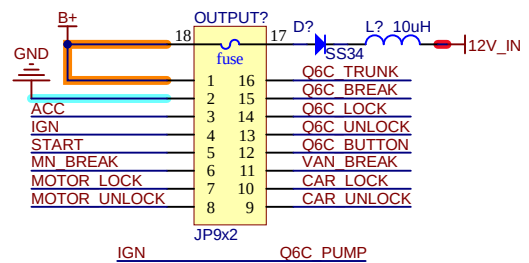
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| Title: *                 |            |              | Author: *             |  |
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| Size: A4                 | Project:   |              | <b>CONFIDENTIAL</b>   |  |
| Modified Date: 8/10/2026 |            |              |                       |  |
| Revision: *              |            |              |                       |  |
| Date: 8/10/2026          | 5:24:36 PM | Sheet 6 of 6 | SW Version: 24.1.2.44 |  |





| Comment          | Description                                                               | Designator | Footprint                          | LibRef          | Quantity |
|------------------|---------------------------------------------------------------------------|------------|------------------------------------|-----------------|----------|
| 1uF              | Cap SMD                                                                   | C?         | C-0603                             | CM              | 1        |
| 10uF             | Cap SMD                                                                   | C?         | C-0603                             | CM              | 4        |
| 22pF             | Cap SMD                                                                   | C?         | C-0603                             | CM              | 2        |
| 100nF            | Cap SMD                                                                   | C?         | C-0603                             | CM              | 7        |
| 100uF/25V        | Polarized Capacitor (Axial)                                               | C?         | CAP-ALU-SMD-D                      | CN-ALU-SMD      | 1        |
| 1N5819           | Default Diode                                                             | D?         | SOD-323                            | Diode SMD       | 3        |
| SS34             | Default Diode                                                             | D?         | DIO-SMD-106                        | Diode SMD       | 3        |
| JP6x1-2.0        | WR-PHD 2.00 mm Socket Header                                              | DEBUGGER?  | J6-D1T                             | JP6x1-2.0       | 1        |
| 10uH             | Inductor                                                                  | L?         | INDUCTOR_SMD_7X7MM                 | Inductor_SMD    | 1        |
| LED              |                                                                           | LED?       | LED-0603-RED                       | LED SMD         | 1        |
| ATFFS150A01BR016 | TFC-9P-1.5H(ATFFS150A01BR016)                                             | MICRO SD?  | USD(TFC-9P-1.XH(ATFFS150A01BR016)) | MICRO_SD        | 1        |
| JP9x2            | WR-PHD 2.54 mm Dual Socket Header                                         | OUTPUT?    | 9x2_JP200                          | JP9x2_1         | 1        |
| AO3401           | P-Channel 20V 2.5A _Tc_ 830mW _Tc_ Surface Mount TO-236AB _SOT23_         | Q?         | SOT23-3                            | SI2303DS        | 2        |
| AO3402           | N-Channel 20V 2.5A _Tc_ 830mW _Tc_ Surface Mount TO-236AB _SOT23_         | Q?         | SOT23-3                            | SI2302DS        | 5        |
| 1K               | Resistor                                                                  | R?         | RES-SMD-0603                       | Res-SMD         | 14       |
| 5.1K             | Resistor                                                                  | R?         | RES-SMD-0603                       | Res-SMD         | 2        |
| 10K              | Resistor                                                                  | R?         | RES-SMD-0603                       | Res-SMD         | 18       |
| SRA-12VDC-AL     | Relay SPDT,10A - 12VDC 5 Pin                                              | RL?        | YL318-DC12V-C                      | Relay 1         | 1        |
| SRA-12VDC-CL     | Relay SPDT,10A - 12VDC 5 Pin                                              | RL?        | YL318-DC12V-C                      | Relay 1         | 2        |
| AP2112K-3.3TRG1  | Linear Voltage Regulator IC Positive Fixed 1 Output 600mA SOT23-5         | U?         | SOT23-5                            | AP2112K-3.3TRG1 | 1        |
| HT7550-1         | Voltage stabiliser; LDO, fixed; 5V; 0.1A; SOT89; SMD                      | U?         | SOT89                              | HT7550-1        | 1        |
| PS2801-4         | LOW INPUT CURRENT, HIGH CTR 4, 16-PIN SSOP PHOTOCOUPLER                   | U?         | SOIC127P700X230-16N                | PS2833-4        | 8        |
| STM32F103C8T6    |                                                                           | U?         | QFP50P900X900X160-48N              | STM32F103C8T6   | 1        |
| 16Mhz            | Crystal - 48MHz Â±10ppm - 10 pF - Â±15 ppm - -30 to +85Â°C 3225 Low aging | Y?         | XTAL_SMD_3325_4P                   | Crystal SMD     | 1        |